STAT lecture 7
One-sample hypothesis tests

Disclaimer:
Please note that these slides are only a brief overview of the topic. Many more details will be explained on the blackboard and with MATLAB examples

SDU

Testing hypotheses for a single random sample (population)

Case 1: Hypothesis test for unknown mean in normal distribution with known variance

Model $X_1, \dots, X_n$ iid $X_i \sim N(\mu, \sigma^2)$, μ unknown, σ^2 known

Null hypothesis

2-sided	H_0: $\mu = \mu_0$	against	H_1: $\mu \neq \mu_0$
1-sided	H_0: $\mu \leq \mu_0$	against	H_1: $\mu > \mu_0$
	H_0: $\mu \geq \mu_0$	against	H_1: $\mu < \mu_0$

Estimate $\hat{\mu} = \overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

Test statistic $Z_0 \stackrel{\text{def}}{=} \frac{\overline{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$ Put in estimate and H_0 → $Z_0 = z_0$

Accept region, A: $z_0 \in A \rightarrow accept\ H_0$

Critical region, C: $z_0 \in C \rightarrow reject\ H_0$

Error types Type 1 error: reject H_0 when H_0 is actually true (most serious error)
Type 2 error: accept H_0 when H_0 is actually false (less serious error)

Level of significance $\alpha \stackrel{\text{def}}{=} max\ acceptable\ P(Type\ 1\ error)$ (Usually 1%, 5% or 10%)

From level of significance, α, to Accept (A) and Reject/Critical regions (C)

2-sided test $H_0: \mu = \mu_0$ against $H_1: \mu \neq \mu_0$

$\alpha = P(z_0 \in C \mid H_0\ true)$

From test statistic pdf $C: |z_0| > z_{\alpha/2}$

$A: |z_0| \leq z_{\alpha/2}$

1-sided test $H_0: \mu \leq \mu_0$ against $H_1: \mu > \mu_0$

$\alpha = P(z_0 \in C \mid H_0\ true)$

From test statistic pdf $C: z_0 > z_\alpha$

$A: z_0 \leq z_\alpha$

$H_0: \mu \geq \mu_0$ against $H_1: \mu < \mu_0$

$\alpha = P(z_0 \in C \mid H_0\ true)$

From test statistic pdf $C: z_0 < -z_\alpha$

$A: z_0 \geq -z_\alpha$

Type 2 error $\beta = \beta(\mu) = P(z_0 \in A \mid H_0\ false)$ (Note: not a number but a function of true μ)

Power $\gamma = \gamma(\mu) = 1 - \beta(\mu) = P(z_0 \in C \mid H_0\ false)$ (Note: not a number but a function of true μ)

p-value (alternative to choosing a fixed level of sigificance, α)

Definition calculate value of test statistic $z_0 \overset{\text{def}}{=} \frac{\bar{X}-\mu_0}{\frac{\sigma}{\sqrt{n}}}$

the p-value is the probability of a value of the test statistic (of the data) at least as rare as the calculated one for the data, GIVEN that H_0 is true

Interpretation if p-value is "small" → reject H_0
optionally compare p-value with usual choices of α

2-sided test $H_0: \mu = \mu_0$ against $H_1: \mu \neq \mu_0$
From test statistic pdf
$p/2 = P(Z > |z_0|)$ → $p = 2\left(1 - P(Z \leq |z_0|)\right) = 2(1 - \Phi(|z_0|))$

1-sided test $H_0: \mu \leq \mu_0$ against $H_1: \mu > \mu_0$
From test statistic pdf
$p = P(Z > z_0)$ → $p = 1 - P(Z \leq z_0) = 1 - \Phi(z_0)$

1-sided test $H_0: \mu \geq \mu_0$ against $H_1: \mu < \mu_0$
From test statistic pdf
$p = P(Z < z_0)$ → $p = \Phi(z_0)$

Case 2: Hypothesis test for unknown mean in normal distribution with unknown variance

Model $X_1, \dots, X_n$ iid $X_i \sim N(\mu, \sigma^2)$, μ unknown, σ^2 unknown

Null hypothesis

2-sided	H_0: $\mu = \mu_0$	against	H_1: $\mu \neq \mu_0$
1-sided	H_0: $\mu \leq \mu_0$	against	H_1: $\mu > \mu_0$
	H_0: $\mu \geq \mu_0$	against	H_1: $\mu < \mu_0$

Estimates

$$\hat{\mu} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\hat{\sigma}^2 = s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 \sim \frac{\sigma^2}{n-1}\chi^2(n-1)$$

Test statistic $T_0 \overset{\text{def}}{=} \frac{\bar{X}-\mu_0}{\frac{s}{\sqrt{n}}} \sim t(n-1)$ Put in estimate and H_0 → $T_0 = t_0$

2-sided test H_0: $\mu = \mu_0$ against H_1: $\mu \neq \mu_0$

$\alpha = P(t_0 \in C \mid H_0\ true)$

From test statistic pdf C: $|t_0| > t(n-1)_{\alpha/2}$

A: $|t_0| \leq t(n-1)_{\alpha/2}$

p-value $p = 2\left(1 - P(t(n-1) \leq |t_0|)\right)$

1-sided test H_0: $\mu \leq \mu_0$ against H_1: $\mu > \mu_0$

$\alpha = P(t_0 \in C \mid H_0\ true)$

From test statistic pdf C: $t_0 > t(n-1)_{\alpha}$

A: $t_0 \leq t(n-1)_{\alpha}$

p-value $p = 1 - P(t(n-1) \leq t_0))$

H_0: $\mu \geq \mu_0$ against H_1: $\mu < \mu_0$

$\alpha = P(t_0 \in C \mid H_0\ true)$

From test statistic pdf C: $t_0 < -t(n-1)_{\alpha}$

A: $t_0 \geq -t(n-1)_{\alpha}$

p-value $p = P(t(n-1) \leq t_0))$

Case 3: Hypothesis test for unknown variance in normal distribution

Model $X_1, \dots, X_n$ iid $X_i \sim N(\mu, \sigma^2)$, μ unknown, σ^2 unknown

Hypothesis $H_0: \sigma^2 = \sigma_0^2$ against $H_1: \sigma^2 \neq \sigma_0^2$ (2-sided test)

Estimate $\hat{\sigma}^2 = s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 \sim \frac{\sigma^2}{n-1}\chi^2(n-1)$

Test statistic $X_0^2 \overset{\text{def}}{=} \frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$ Put in estimate and H_0 → $X_0^2 = x_0^2$

Test From test statistic pdf

C: $x_0^2 > \chi^2(n-1)_{\alpha/2} \vee x_0^2 < \chi^2(n-1)_{1-\alpha/2}$

A: $\chi^2(n-1)_{1-\alpha/2} \leq x_0^2 \leq \chi^2(n-1)_{\alpha/2}$

p-value $p = 2\,min\{P(\chi^2(n-1) > x_0^2), P(\chi^2(n-1) < x_0^2)\}$

Case 4: Hypothesis test for succes rate in binomial distribution

Model $X \sim Bin(n, p)$ H_0: $p = p_0$

Use large sample normal distribution approximation of binomial and Z-test (case 1)